muscles than when you laugh because something is really funny. Humans cannot usually detect the difference. But a biometric sensor could.[9]

The word television comes from Greek '*tele*', which means 'far', and Latin '*visio*', sight. It was originally conceived as a device that allows us to see from afar. But soon, it might allow us to *be seen* from afar. As George Orwell envisioned in *Nineteen Eighty-Four*, the television will watch us while we are watching it. After we've finished watching Tarantino's entire filmography, we may have forgotten most of it. But Netflix, or Amazon, or whoever owns the TV algorithm, will know our personality type, and how to press our emotional buttons. Such data could enable Netflix and Amazon to choose movies for us with uncanny precision, but it could also enable them to make for us the most important decisions in life – such as what to study, where to work, and who to marry.

Of course Amazon won't be correct all the time. That's impossible. Algorithms will repeatedly make mistakes due to insufficient data, faulty programming, muddled goal definitions and the chaotic nature of life.[10] But Amazon won't have to be perfect. It will just need to be better on average than us humans. And that is not so difficult, because most people don't know themselves very well, and most people often make terrible mistakes in the most important decisions of their lives. Even more than algorithms, humans suffer from insufficient data, from faulty programming (genetic and cultural), from muddled definitions, and from the chaos of life.

You may well list the many problems that beset algorithms, and conclude that people will never trust them. But this is a bit like cataloguing all the drawbacks of democracy and concluding that no sane person would ever choose to support such a system. Winston Churchill famously said that democracy is the worst political system in the world, except for all the others. Rightly or wrongly, people might reach the same conclusions about Big Data algorithms: they have lots of hitches, but we have no better alternative.

As scientists gain a deeper understanding of the way humans make decisions, the temptation to rely on algorithms is likely to increase. Hacking human decision-making will not only make Big Data algorithms more reliable, it will simultaneously make human feelings *less* reliable. As governments and corporations succeed in hacking the human operating system, we will be exposed to a barrage of precision-guided manipulation, advertisement and propaganda. It might become so easy to manipulate our opinions and emotions that we will be forced to rely on algorithms in the same way that a pilot suffering an attack of vertigo must ignore what his own senses are telling him and put all his trust in the machinery.

In some countries and in some situations, people might not be given any choice, and they will be forced to obey the decisions of Big Data algorithms. Yet even in allegedly free societies, algorithms might gain authority because